Why algorithms

Extract device/active nodes from poly/ndiff/pdiff

The idea

- GOLDEN: extract_nodes=3

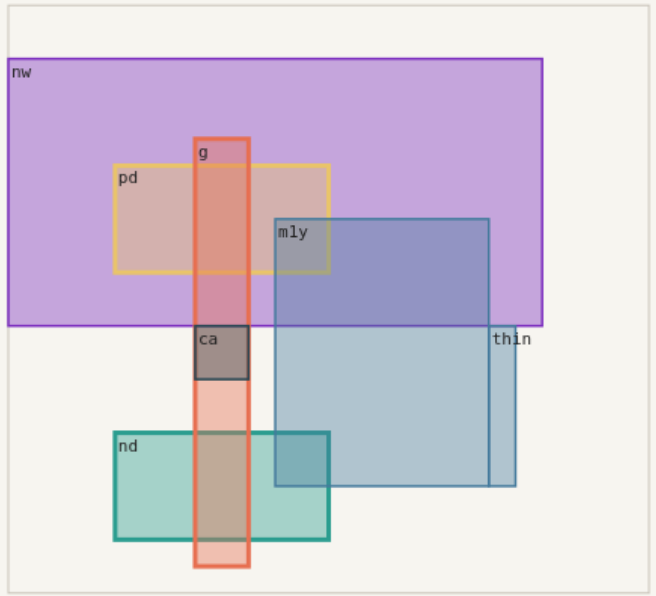
Gate poly

EXTRACT ALGORITHM

Gate poly

Step 1 / 5 · poly · module10-01-extract-algo

poly



Explanation

Extract poly shape g as gate node.

- g

```
poly
```

Gate poly

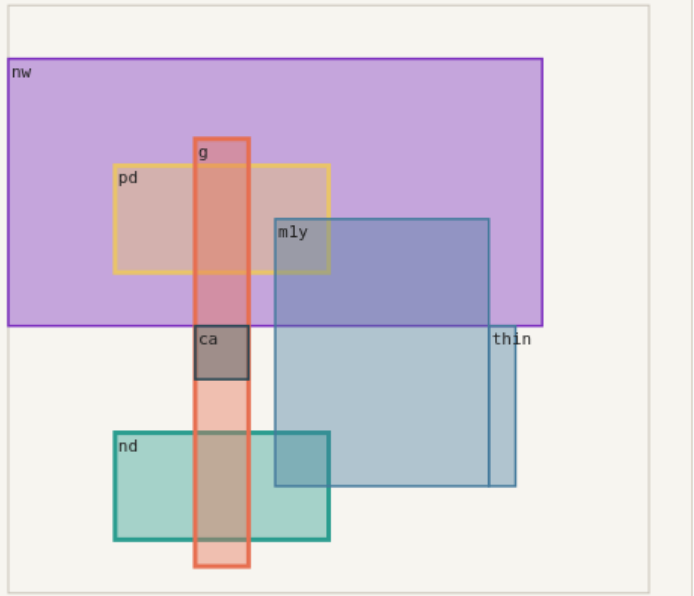
ndiff

EXTRACT ALGORITHM

ndiff

Step 2 / 5 · nd · module10-01-extract-algo

ndiff



Explanation

Extract nd as source/drain region.

- nd

```
ndiff
```

ndiff

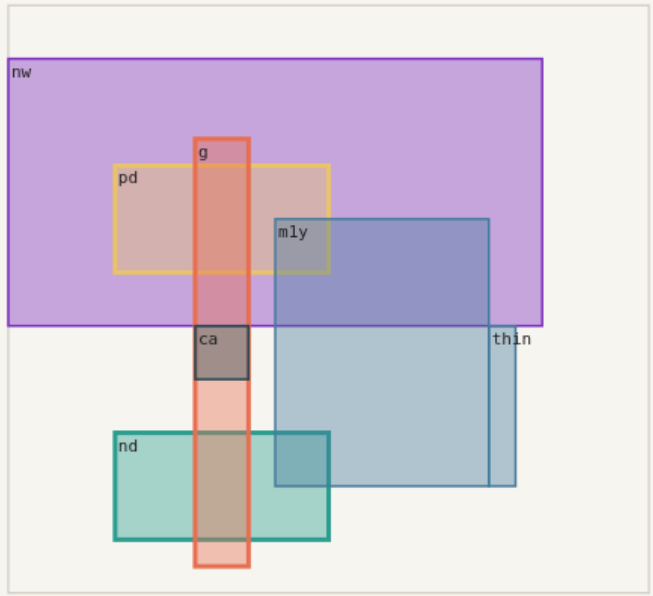
pdiff

EXTRACT ALGORITHM

pdiff

Step 3 / 5 · pd · module10-01-extract-algo

pdiff



Explanation

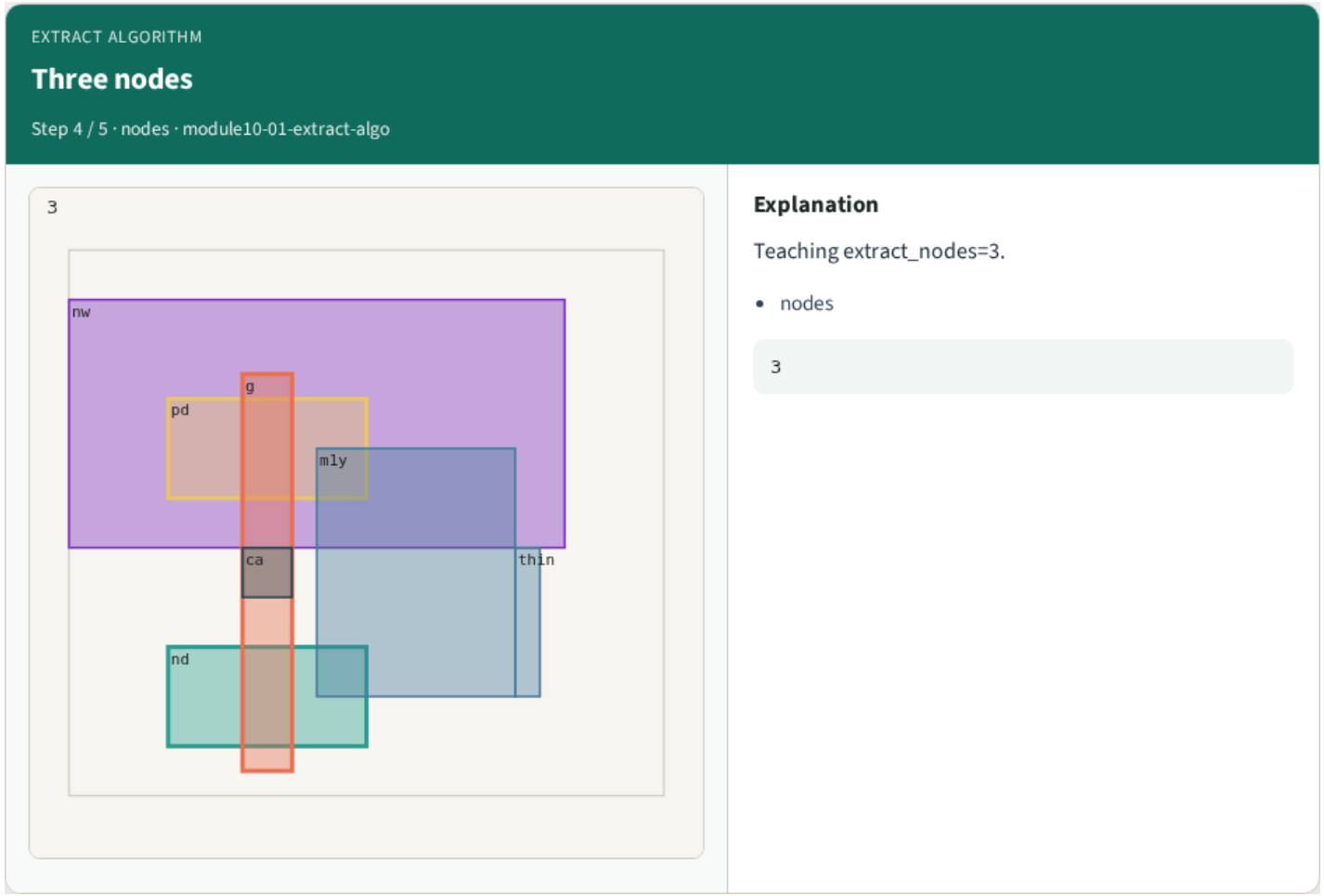
Extract pd as source/drain region.

- pd

```
pdiff
```

pdiff

Three nodes



Three nodes

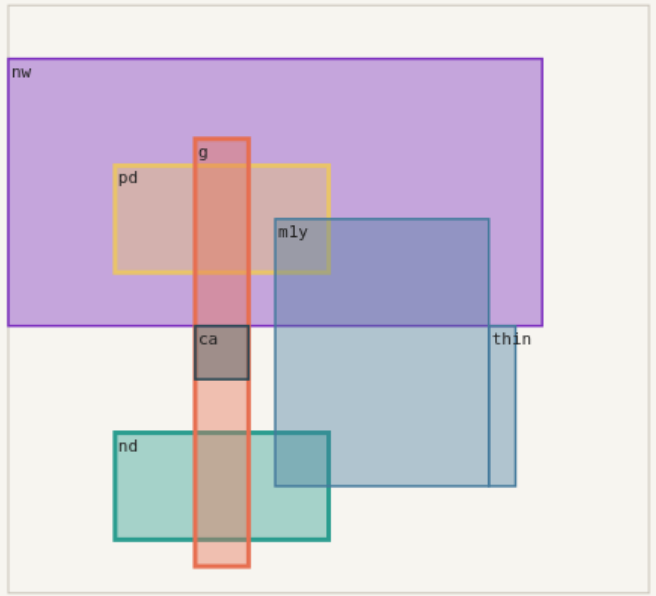
Golden

EXTRACT ALGORITHM

Golden

Step 5 / 5 · golden · module10-01-extract-algo

3



Explanation

GOLDEN: extract_nodes=3.

- GOLDEN

```
3
```

Golden

Browser lab track

- Open **extract-algo**
- Run connectivity / DRC / extract / bind, then Check

Implement track

- From common/, run algo_solvers.py and partc_bind.py

Pitfalls

- Changing tiny_algo_magic without updating Part B goldens
- Ignoring the bind side

Try these

- From the common folder, run the Part C self-tests

Try these — try these

```
cd courses/learn_magic/common
python3 algo_solvers.py
python3 partc_bind.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
extract_nodes=3.
ok if goldens match and bind
```

Next

- Complete the quiz for this part
- Continue Part C, then internals literacy